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Multi-objective optimization of an EDM process for Monel K-500 alloy using response surface methodology-multi-objective dragonfly algorithm

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Monel K-500 is a high-performance superalloy composed of nickel and copper, renowned for its exceptional strength, hardness, and resistance to corrosion. To machine this material more precisely and accurately, Electrical Discharge Machining (EDM) is one of the best choices. In EDM, material removal rate (MRR) and electrode wear rate (EWR) are crucial performance parameters that are often conflicting in nature. These parameters depend on several input variables, including peak current (Ip), pulse on time (Ton), duty cycle (Tau), and servo voltage (SV). Optimizing the EDM process is essential for enhancing performance. In this research, a set of experiments were conducted using EDM on Monel K500 alloy to determine the optimal process parameters. The Box–Behnken design was used to prepare the experimental design matrix. Utilizing the experimental data, a second-order mathematical model was developed using Response Surface Methodology (RSM). R^2 value is found to be 99.40% and 96.60% for MRR and EWR RSM-based prediction model, respectively. High value of R^2 is indicated is indicated good adequacy for prediction. The mathematical model further used in multi-objective dragonfly algorithm (MODA): a new meta-heuristic optimization technique to solve multi-objective optimization problem of EDM. The MODA is a very useful technique to achieve optimal solutions from the multi decision criteria. Utilizing this technique, a set of non-dominated solutions was obtained. Further, the TOPSIS method was used to determine the most desirable optimal solution, which was found to be 0.0135 mm^3/min for EWR and 6.968 mm^3/min for MRR. These results were obtained when the optimal process parameters were selected as Ip = 6 A, Ton = 200 µs, Tau = 12, and SV = 41.6 V. Operators can machine Monel K500 by selecting the above-mentioned optimal parameters to achieve the best performance.

Keywords EDM, Monel K-500, RSM, MODA

Currently, manufacturing industries face the challenge of machining various alloy or composite materials with high strength, hardness, and temperature resistance. Unconventional machining processes are employed to handle such difficult-to-cut materials, ensuring high surface finish, precise dimensional accuracy, and intricate shapes.[1]. Electro-discharge machining (EDM) employs the thermal energy of sparks to remove material. It finds application in the machining of hardened steel dies, aerospace, automotive, and machine tool components, as well as in the production of medical components[2,3]. The extensive use of EDM in manufacturing, particularly for machining newly developed advanced materials, renders it a significant subject of research. Numerous researchers have delved into the EDM process while machining various materials. Ahmed et al.[4] investigate the EDM process with the aim of minimizing geometrical errors, surface roughness, and tool wear in machined titanium alloy components. Asif et al.[5] found that machining efficiency is a crucial indicator for sustainable EDM machining.

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To improve efficiency, eco-friendly Tween series surfactants were added to the dielectric fluid while machining Ti6Al4V ELI alloy and investigated the effects of various process parameters on performance. The material removal rates (MRR) and tool wear rates (TWR) of EDM are crucial performance parameters[6–8]. Farooq et al.[9] identified and investigated the influence of process parameters, namely pulse current, pulse on time, pulse off time, polarity, and dielectric, on EDM performance during the machining of titanium alloy. Various researchers observed that different process parameters have diverse effects on multiple performance parameters. Nguyen et al.[10] observed a significant impact of peak current on machinability, noting that crater size depends on spark energy. Balasubramanian et al.[11] found that peak current is the principal influencing parameter for MRR and surface roughness during the machining of high manganese steel. During machining titanium alloy, the pulse on time emerges as a predominant factor[12]. Izwan et al.[13] utilized four different materials: brass, aluminum, high-strength steel, and high-strength low-alloy steel. It is found that higher peak current and longer pulse-on time resulted in an increased Material Removal Rate (MRR). Tran et al.[14] optimized the machining process of AISI P20 steel using Taguchi and ANOVA analyses. It is highlighted that current significantly impacts Material Removal Rate (MRR), Electrode Wear Rate (EWR), and Surface Roughness (SR). The surface roughness found to be decrease with increases in pulse on time, pulse off time, and current. Hussain et al.[15] assessed that peak current is the primary determinant factor that affecting on MRR and EWR during the machining of aluminum oxide-copper composite, by employing the Taguchi method. Researchers have established that different EDM performance parameters are influenced by different process parameters such as pulse-on time, pulse-off time, peak current, and voltage.

The performance parameters in EDM often conflict, with different process variables impacting various outcomes in diverse ways. Therefore, optimally selecting these parameters is crucial for enhancing EDM performance. To achieve this, a range of modeling and optimization techniques have been extensively employed to identify the most suitable machining parameters, thereby improving overall performance[16–18]. To enhance the sustainability of the EDM process, Sana et al.[19] used alumina-mixed deionized water as the dielectric fluid. Additionally, the EDM process was modeled using artificial neural networks (ANN) and optimized it with non-dominated sorting genetic algorithms (NSGA-II). Machine learning based predictive model for EDM process found to be an effective way to enhancing the performance[20,21]. Kaigude et al.[22] employed machine learning methods, including linear regression, decision trees, and random forests, to predict the surface roughness during the machining of AISI D2 steel in the presence of Titanium dioxide (TiO2) nanopowder in the dielectric. Seidi et al.[23] applied methods based on the removal effects of criteria (MEREC) and the weighted aggregates sum product assessment (WASPAS) techniques to address multi-objective optimization in the wire electrical discharge machining process. Sing et al.[24] employed Meta-heuristic optimization techniques, including Teaching Learning-Based Optimization (TLBO) and Particle Swarm Optimization (PSO) algorithms, for electro-discharge machining of 316L porous stainless steel. Mandal and Mondal[25] used MOPSO-TOPSIS to solve the multi objective optimization problem of EDM. Bhowmick et al.[26] developed a prediction model for Material Removal Rate (MRR) and surface roughness in titanium-mixed Electrical Discharge Machining (EDM) of Inconel 718 using Response Surface Methodology (RSM) and fuzzy logic, optimizing the process parameters. RSM encompasses a set of mathematical and statistical techniques valuable for modeling and analyzing problems where a response of interest is affected by multiple variables, with the aim of optimizing these responses[27]. Joshi et al.[28] compared multi-objective optimization techniques, including the non-dominated sorting genetic algorithm II (NSGA-II), multi-objective ant lion optimization (MOALO), and multi-objective dragonfly optimization (MODA), in micro-turning and micro-milling. Chang et al.[29] employed the NSGA-II algorithm to solve a multi-objective optimization problem. Wang et al.[30] utilized MODA analysis to develop a hybrid forecasting framework in electrical power systems. While these meta-heuristic techniques have effectively addressed optimization problems, their application in solving multi-objective optimization problems in EDM operations is rare.

Monel K-500, a nickel-based superalloy, exhibits exceptional corrosion resistance, as well as high strength and toughness across a broad temperature range[31].

Machining nickel-based alloys such as Monel K-500 poses a challenge for many traditional machining processes due to their inherent limitations. EDM is a correct alternative solution for machining of Monel materials. In a study conducted by Akgün[32], machining of Monel K-500 superalloy using Electrical Discharge Machining (EDM) was investigated with various electrodes. The results indicate that the copper electrode outperforms the graphite electrode for machining Monel K-500 alloy. Though the nickel-based alloy Monel K-500 has huge applicability and EDM is an effective way for machining, very few research works are found on experimental investigation and optimization of the EDM process during machining of Monel K500 material. Therefore, a number of experiments were conducted adopting RSM Box-Behnken design of experiment on Monel K-500 alloy. Mathematical models were developed using RSM for MRR and EWR. Further, the RSM models were used as objective function in dragonfly algorithm: a new meta-heuristic optimization technique to solve the multi objective optimization problem.

The extensive literature survey has highlighted that the nickel-based alloy Monel K-500 is difficult to machine using traditional machining processes. However, non-traditional methods, such as Electrical Discharge Machining (EDM), present a viable alternative. It was also found that different process parameters have varying effects on performance parameters, with many performance parameters exhibiting conflicting behaviors. Modeling and optimization are essential techniques for enhancing EDM performance. Therefore, in this current work, EDM operations were performed on Monel K-500 alloy to solve a multi-objective optimization problem. To determine the optimal EDM process parameters, Response Surface Methodology (RSM) and a newly developed Multi-Objective Dragonfly Algorithm (MODA) were employed.

Materials and methods

Materials

After conducting a comprehensive review of the existing literature on the parametric optimization of Electrical Discharge Machining (EDM) parameters for various materials and alloys, MONEL K-500 alloy was selected as the workpiece material for experimentation. Tables 1 and 2 provide the chemical composition and physical properties of the MONEL K-500 alloy, respectively. As shown in Table 2, the mechanical properties of MONEL K-500, such as yield strength, ultimate tensile strength, and hardness, are relatively high, making it challenging to machine using traditional methods. However, its high electrical conductivity makes it suitable for EDM processes.

A flat rectangular plate with dimensions of (116 × 75) mm and a thickness of 5 mm was used for the experiments. The experiments were conducted following the Box–Behnken design of Response Surface Methodology.

Experimental setup

The flowchart illustrating the current research is shown in Fig. 1. The experiments were conducted using a die-sinking EDM machine (Model ELTECH D-300ZNC, India) at the IIEST, Shibpur, India, as illustrated in Fig. 2. Since the workpiece material is non-magnetic, the plate was secured in place by clamping it with mild steel plates, which are magnetic, as depicted in Fig. 3. Blind holes with a depth of 0.5 mm were created for each set of parameters. To calculate the material removal rate (MRR) and electrode wear rate (EWR), the time taken during each machine run was carefully monitored using a stopwatch and recorded. The weight before and after machining each specimen was measured using a precision weighing machine with a least count of 0.001g.

Element	Ni + Co	Cu	Fe	Al	Ti	Mn
Concentration (weight %)	63% min	27–33%	2% max	2.3–3.15% max	0.35–0.85% max	0–1.5% max

Table 1. Chemical composition of Monel K-500 alloy.

Properties	Units	Value
Density	g/cm^3	8.44
Yield strength	MPa	690
Ultimate tensile strength	MPa	965
Hardness	Brinell (3000-kg)	250–315
Thermal conductivity	W/m K	17.5
Electrical resistivity	μΩ-cm	6.15
Melting point	°C	1350

Table 2. Physical properties of MONEL K-500 Alloy[32].

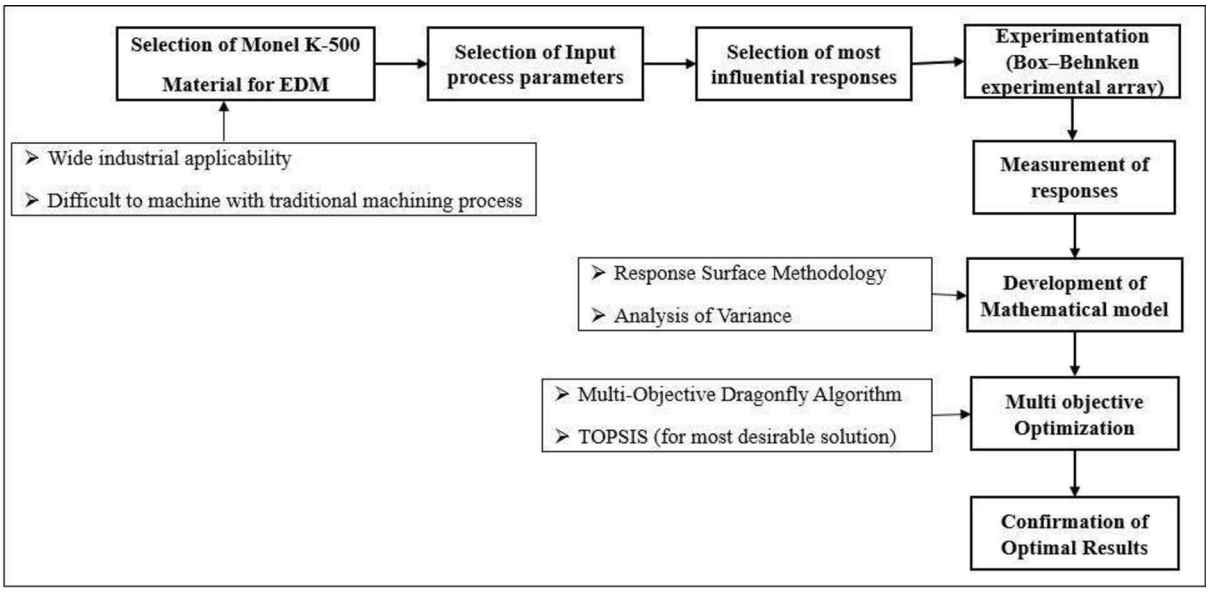

Fig. 1. Flowchart of research.

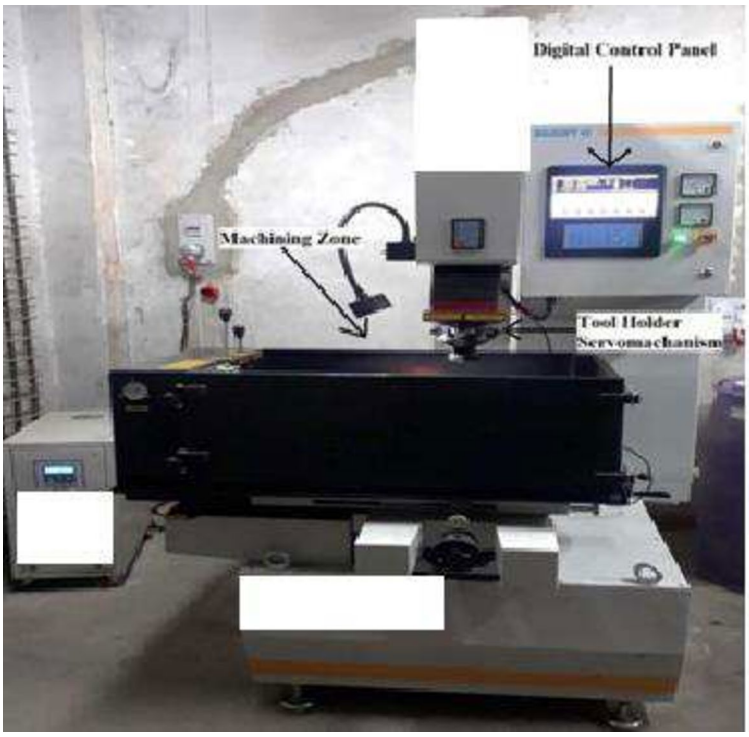

Fig. 2. EDM experimental set-up.

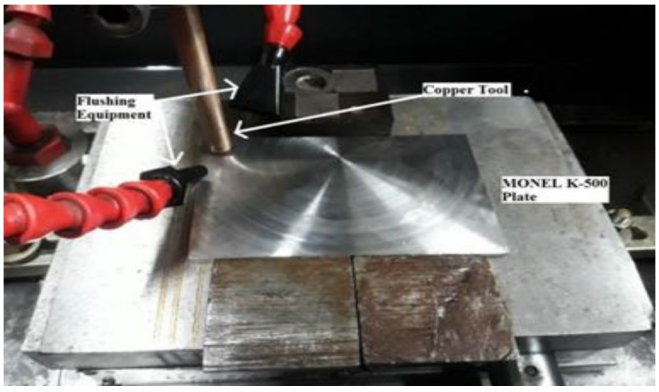

Fig. 3. Pre-machining setup.

Experimental array

The experimental array is formed by combining input process variables to determine the conditions under which the experiments are conducted. This array is influenced by different process variables and their respective settings. In this particular study, four process variables—peak current (Ip), pulse on time (Ton), duty cycle (Tau), and servo voltage (SV)—are employed for experimental purposes. These parameters significantly affect the Material Removal Rate (MRR) and Electrode Wear Rate (EWR). In EDM, optimizing these parameters is crucial for achieving the desired balance between MRR and EWR. While higher peak currents, pulse-on times, and duty cycles increase MRR, they also tend to increase EWR. Conversely, higher servo voltages decrease both MRR and EWR by widening the spark gap and reducing discharge intensity. Therefore, finding an optimal set of parameters is essential for efficient and effective machining, ensuring high MRR while minimizing electrode wear. The range of input machining variables is detailed in Table 3. The selection of process variables, their ranges, and levels is based on the experimental data set provided with EDM.

In the current research, a Box–Behnken-based experimental array is employed within an identified search space (n = 1). The initial investigation aimed to confine the input process variables within the operational range. Following the determination of this range, the experimental design was executed, as outlined in Table 4. All experiments were conducted according to the run order rather than a conventional sequence. This approach aligns with the principle of randomness, ensuring the reproducibility of machine tool results. A total of 27

Level	Ip (Amps)	Ton (µs)	Tau	SV (V)
1	6	100	12	40
2	8	150	14	45
3	10	200	16	50

Table 3. EDM Parameters and Levels.

Exp. No.	Ip (Amps)	Ton (µs)	Tau	SV (V)	MRR (mm^3/min)	EWR (mm^3/min)
1	8	100	14	40	14.48869	0.26761
2	10	100	14	45	21.30935	0.53706
3	8	150	12	50	11.83526	0.21034
4	6	100	14	45	8.65506	0.08319
5	8	200	12	45	10.33148	0.07121
6	6	150	16	45	8.518809	0.0757
7	8	150	14	45	13.01855	0.1858
8	6	200	14	45	7.56395	0.04582
9	8	150	16	50	15.27399	0.16443
10	8	150	14	45	13.43348	0.1883
11	8	150	12	40	11.51796	0.18623
12	10	150	14	50	19.96366	0.33819
13	8	100	12	45	13.16322	0.36576
14	6	150	14	50	7.85814	0.07773
15	10	200	14	45	17.40882	0.26971
16	8	100	14	50	15.39212	0.23712
17	10	150	16	45	22.3306	0.32449
18	8	200	14	40	12.37341	0.07042
19	8	200	14	50	11.54403	0.06692
20	6	150	14	40	7.51217	0.07783
21	6	150	12	45	7.04428	0.07868
22	8	200	16	45	12.59403	0.06384
23	8	150	16	40	14.36706	0.17949
24	8	150	14	45	13.16962	0.17978
25	10	150	14	40	20.3209	0.33808
26	10	150	12	45	18.2627	0.3333
27	8	100	16	45	17.75811	0.21583

Table 4. Experimental parameters and machining responses.

experiments were carried out, with each experiment being replicated twice to uphold the statistical precision of the results.

Response surface methodology

Mathematical modelling using Response Surface Methodology (RSM)[33,34] involves developing mathematical equations to represent the relationship between input factors (independent variables) and a response (dependent variable). The primary goal is to create a predictive model that can guide experimentation and optimization of the system.

Here are the key steps in mathematical modelling using RSM:

Experimental design:
Conduct a well-planned experimental design, varying the input factors at different levels. Use a factorial design or fractional factorial design to efficiently explore the factor space.
Data collection:
Collect data on the response variable at each combination of factor levels.
Ensure that the data collection is accurate and representative of the system under study.
Fit a mathematical model:
Choose a suitable mathematical model based on the nature of the relationship between factors and response.
Common models include linear, quadratic, and cubic equations.

The general quadratic model can be represented as:

$$Y = b_0 + \sum_{i=1}^{k} b_i X_i + \sum_{i=1}^{k} b_{ii} X_i^2 + \sum_{i=1}^{k-1} \sum_{j=i+1}^{k} b_{ij} X_i X_j + \epsilon \quad (1)$$

where: Y is the predicted response, $b_0, b_i, b_{ii} and b_{ij}$ are coefficients to be determined, X_i represents the levels of the independent variables, ϵ is the error term.
Parameter estimation:
Use statistical methods such as least squares estimation to determine the coefficients in the model.
Model validation:
Validate the model by comparing predicted responses with actual experimental data not used in the model fitting. Statistical techniques such as analysis of variance (ANOVA) are often employed for model validation.

Multi-Objective Dragonfly Algorithm (MODA)

The Dragonfly Algorithm (DA) is a nature-inspired optimization algorithm based on the swarming behaviour of dragonflies. It is a meta-heuristic optimization technique that simulates the social interactions and foraging behaviour of dragonfly swarms to solve optimization problems. Mirjalili[35] introduced the MODA, an optimization algorithm founded on swarm intelligence, in 2014. Here's an overview of the Dragonfly Optimization Algorithm:

- Swarming Behaviour:
  The algorithm is inspired by the collective behaviour of dragonflies in nature, where they exhibit coordinated movements and group hunting for efficient prey capture.
- Search Agents (Dragonflies):
  Dragonflies in the algorithm represent the search agents. Each dragonfly corresponds to a potential solution in the search space.
- Objective Function:
  The optimization problem is defined by an objective function that needs to be either minimized or maximized.
- Movement and Interaction:
  Dragonflies move within the search space based on their current positions and the positions of other dragonflies. This movement is influenced by social interactions.
- Prey Capture and Exploration:
  Dragonflies engage in prey capture behaviour, focusing on regions with promising solutions. Exploration and exploitation are balanced to avoid premature convergence.
- Algorithm Steps:

  Initialization:
  Initialize a population of dragonflies with random positions in the search space.
  Evaluation:
  Evaluate the objective function for each dragonfly to determine their fitness.
  Movement:
  Update the position of each dragonfly based on its current position, the positions of other dragonflies, and predefined movement rules.
  Prey Capture:
  Dragonflies adjust their positions to focus on areas with better solutions, mimicking the prey capture behaviour in nature.
  Update Best Solution:
  Update the global best solution if a dragonfly discovers a better solution than the current best.
  Termination:
  Repeat the movement and prey capture steps iteratively until a stopping criterion is met (e.g., a maximum number of iterations or achieving a satisfactory solution).

The Dragonfly Algorithm has been applied to various optimization problems, including engineering design, scheduling, and parameter optimization in machine learning.

Results and discussion

The regression model employing Response Surface Methodology (RSM) has been established to predict Material Removal Rate (MRR) and Electrode Wear Rate (EWR). This model is formulated as a function of peak current (Ip), pulse on time (Ton), duty cycle (Tau), and servo voltage (SV), utilizing experimental data. The adequacy of the developed quadratic models was evaluated through Analysis of Variance (ANOVA). The significance of both the overall model and individual model terms was determined using F-tests and P-tests.

Influence of process parameters on MRR

ANOVA is utilized to assess the significance and percentage contribution of all elements in the model. Backward elimination is applied to eliminate insignificant terms without influence on the model, and the model's adequacy is tested at each step. The terms with p values less than 0.05 are considered significant. Table 4 shows the ANOVA

and fit summary of MRR during machining of Monel K500 using EDM. Linear term SV, square term Ton × Ton, Tau × Tau and 2-Way Interaction term Ip × SV, Ton × SV and Tau × SV are found to be insignificant as p-value greater than 0.05 and this term can be removed from the model using backward elimination. Table 5 shows the R^2 value for the model. R^2 is an important statistical parameter which defines the variability in responses. Higher value of R^2 shows the good correlation between the response values and experimental values. Moreover, R^2 increases while adding terms to the model. It does not predict whether the added terms are significant or insignificant. Thus, the fitness of the regression model cannot be explained by a larger R^2. Therefore, another statistical parameter namely adjusted R^2(R^2-adj) is used which decreases by the inclusion of insignificant terms to the model. The R^2 value is found to be 99.40% which implies the high relational factor between variables and factors.

Figure 4 displays four distinct plots: a normal probability plot illustrating the relationship between residuals and percent, versus fit plots portraying the relationship between fitted values and residuals, a histogram depicting the frequency distribution of residuals, and order plots illustrating the relationship between observation order and residuals. These plots are presented for MRR. In Fig. 3, it is observed that the residual values, representing the differences between experimental and mathematically predicted values, closely align with the normal probability line. This observation suggests that the errors are distributed in a normal manner, indicating that the models are suitable for prediction. The residuals for the MRR prediction model range from -5 to 5. Lower values within these ranges indicate higher accuracy in the RSM prediction models. In Fig. 3, a notable concentration of frequency is observed around the zero value, indicating minimal error in mathematical modelling within the RSM model. The graphical representation of residuals (the difference between experimental and mathematically predicted values) versus observed values is shown in Fig. 3. The residuals appear randomly scattered around zero across various observed values, indicating a well-fitted RSM model for prediction.

Percentage contribution (PC) of terms are depicted in same ANOVA table (Table 4). It is found that in linear part, Ip is the dominating EDM parameter with PC of 86.16 followed by Ton and T_{au} are 5.9 and 5.73, respectively. Spark current (Ip) has highest influence on MRR. High current supply means high energy supply to the sparking zone which causes higher amount of material remove from machining zone. The empirical quadratic mathematical model, expressed in coded units for MRR as a function of peak current (Ip), pulse on time (Ton), duty cycle (Tau), and servo voltage (SV), is presented in Eq. (2).

Source	DF	Adj SS	Adj MS	F-value	P-value	% contribution
Model	14	504.562	36.04	143.13	0	99.40
Linear	4	496.508	124.127	492.96	0	97.82
Ip	1	437.34	437.34	1736.86	0	86.16
Ton	1	29.928	29.928	118.86	0	5.90
Tau	1	29.103	29.103	115.58	0	5.73
SV	1	0.138	0.138	0.55	0.473	0.03
Square	4	2.078	0.519	2.06	0.149	0.41
Ip*Ip	1	1.912	1.912	7.59	0.017	0.38
Ton*Ton	1	0.032	0.032	0.13	0.726	0.01
Tau*Tau	1	0.09	0.09	0.36	0.561	0.02
SV*SV	1	0.02	0.02	0.08	0.783	0.00
2-way interaction	6	5.976	0.996	3.96	0.02	1.18
Ip*Ton	1	1.973	1.973	7.84	0.016	0.39
Ip*Tau	1	1.681	1.681	6.68	0.024	0.33
Ip*SV	1	0.124	0.124	0.49	0.497	0.02
Ton*Tau	1	1.36	1.36	5.4	0.038	0.27
Ton*SV	1	0.751	0.751	2.98	0.11	0.15
Tau*SV	1	0.087	0.087	0.35	0.568	0.02
Error	12	3.022	0.252			0.60
Lack-of-fit	10	2.933	0.293	6.65	0.138	0.58
Pure error	2	0.088	0.044			0.02
Total	26	507.583				
Model summary						
S	**R-sq**	**R-sq (adj)**	**R-sq (pred)**			
0.501796	99.40%	98.71%	96.63%			

Table 5. Analysis of variance-MRR.

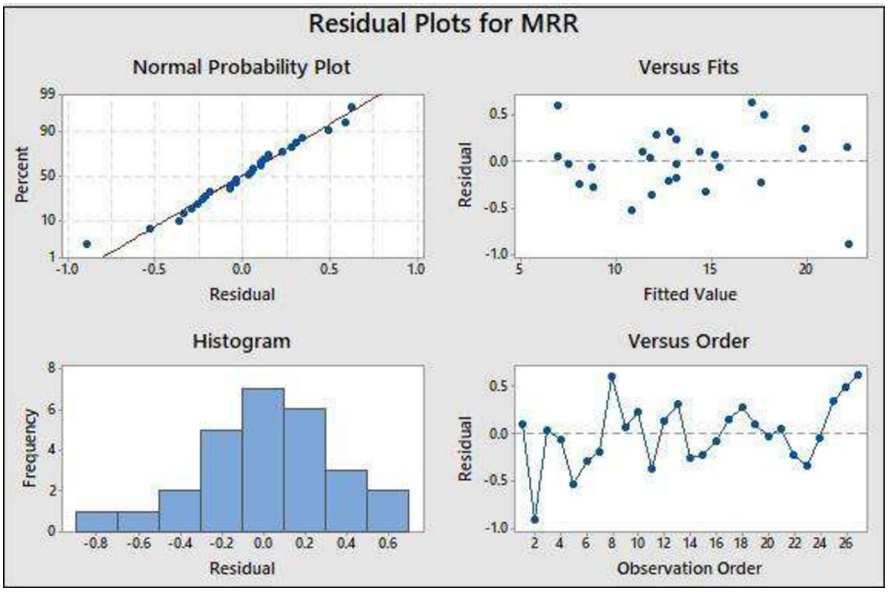

Fig. 4. Residual plots of RSM model for MRR.

Source	DF	Adj SS	Adj MS	F-value	P-value	% contribution
Model	14	0.379002	0.027072	24.32	0	96.59621212
Linear	4	0.349797	0.087449	78.57	0	89.1527359
Ip	1	0.241366	0.241366	216.87	0	61.51693483
Ton	1	0.104281	0.104281	93.7	0	26.57809087
Tau	1	0.004097	0.004097	3.68	0.079	1.044202091
SV	1	0.000052	0.000052	0.05	0.833	0.013253236
Square	4	0.010327	0.002582	2.32	0.116	2.632041737
Ip*Ip	1	0.006609	0.006609	5.94	0.031	1.684435348
Ton*Ton	1	0.000001	0.000001	0	0.975	0.00025487
Tau*Tau	1	0.000069	0.000069	0.06	0.807	0.017586025
SV*SV	1	0.000585	0.000585	0.53	0.482	0.149098907
2-way interaction	6	0.018878	0.003146	2.83	0.059	4.811434484
Ip*Ton	1	0.013223	0.013223	11.88	0.005	3.370145046
Ip*Tau	1	0.000008	0.000008	0.01	0.932	0.002038959
Ip*SV	1	0	0	0	0.998	0
Ton*Tau	1	0.005081	0.005081	4.57	0.054	1.2949941
Ton*SV	1	0.000182	0.000182	0.16	0.693	0.046386327
Tau*SV	1	0.000384	0.000384	0.34	0.568	0.097870052
Error	12	0.013356	0.001113			3.404042747
Lack-of-fit	10	0.013317	0.001332	69.43	0.014	3.39410282
Pure error	2	0.000038	0.000019			0.009685057
Total	26	0.392357				
Model summary						
S	**R-sq**	**R-sq(adj)**	**R-sq(pred)**			
0.033361	96.60%	92.62%	80.43%			

Table 6. Analysis of variance-EWR.

$$\begin{aligned} MRR = & -7.7 + 0.20 \times Ip + 0.1748 \times Ton - 1.22 \times Tau - 0.005 \times SV + 0.1497 \times Ip \times Ip \\ & + 0.000031 \times Ton \times Ton + 0.0325 \times Tau \times Tau + 0.00245 \times SV \times SV \\ & - 0.00702 \times Ip \times Ton + 0.1621 \times Ip \times Tau - 0.0176 \times Ip \times SV - 0.00583 \times Ton \times Tau \\ & - 0.00173 \times Ton \times SV + 0.0147 \times Tau \times SV \end{aligned} \tag{2}$$

Influence of process parameters on EWR

A full quadratic model, similar to the MRR analysis, also investigates for EWR of EDM operation. ANOVA table (Table 6) shows that linear term Tau and SV, square terms, Ton × Ton, Tau × Tau and SV × SV, and two-way interaction term Ip × Tau, Ip × SV, Ton × SV and Tau × SV are found to be insignificant. Linear term Ip has highest percentage of contribution with 61.51% on EWR and it is followed by another linear term Ton with percentage of contribution of 26.57%. The R^2 value for the EWR model is found to be 96.60% which implies the high adequacy for prediction of EWR. Residual plot (Fig. 5) for EWR is supporting evidence for the adequacy of the predicted model. From the residual plots, it is noticed that p value calculated based on Anderson–Darling (AD) statistic test is greater than the significance level of 0.05. This indicates that the residuals are normally distributed, and it can be inferred that the developed quadratic model for EWR is adequate. The residuals for the EWR prediction model range from − 0.05 to 0.05. Lower values within these ranges indicate higher accuracy in the RSM prediction models. The histogram plot depicting the frequency distribution of residuals. From the plot, it is observed that residual is concentrated on very close to zero. All these evidence are indicated a very high adequacy of EWR model. The empirical quadratic mathematical model, expressed in coded units for EWR as a function of peak current (Ip), pulse on time (Ton), duty cycle (Tau), and servo voltage (SV), is presented in Eq. (3).

$$\begin{aligned} EWR = & -0.84 + 0.021 \times Ip - 0.00341 \times Ton + 0.010 \times Tau + 0.0469 \times SV \\ & + 0.00880 \times Ip^2 - 0.000000 \times Ton^2 - 0.00090 \times Tau^2 - 0.000419 \times SV^2 \\ & - 0.000575 \times Ip \times Ton - 0.00036 \times Ip \times Tau + 0.00001 \times Ip \times SV \\ & + 0.000356 \times Ton \times Tau + 0.000027 \times Ton \times SV - 0.00098 \times Tau \times SV \end{aligned} \tag{3}$$

Multi-objective optimization using MODA

The key EDM responses, EWR and MRR, exhibit conflicting characteristics.

The important EDM response EWR and MRR are conflicting in nature. Different process parameters have different percentages of contribution on EWR and MRR. Therefore, it is crucial to identify the optimal combination of input parameters to achieve reduced EWR and increased MRR. Below are two objective functions provided:

$$\text{Objective 1} = \text{Minimize (MRR)}$$
$$\text{Objective 2} = \text{Maximize (EWR)}$$

In this research, MODA was utilized for multi-objective optimization under parameter constraints. The fitness functions in MODA (MATLAB 2015a) are based on a second-order mathematical model for EWR "(1)" and MRR "(2)". The population size, or number of search agents, is maintained at 100. Equal weightage is assigned to each fitness function. To minimize EWR and maximize MRR, specific boundary conditions should be adhered to selecting the EDM process parameters, namely peak current (Ip), pulse on time (Ton), duty cycle (Tau), and servo voltage (SV). The boundary conditions for the process parameters are outlined as follows:

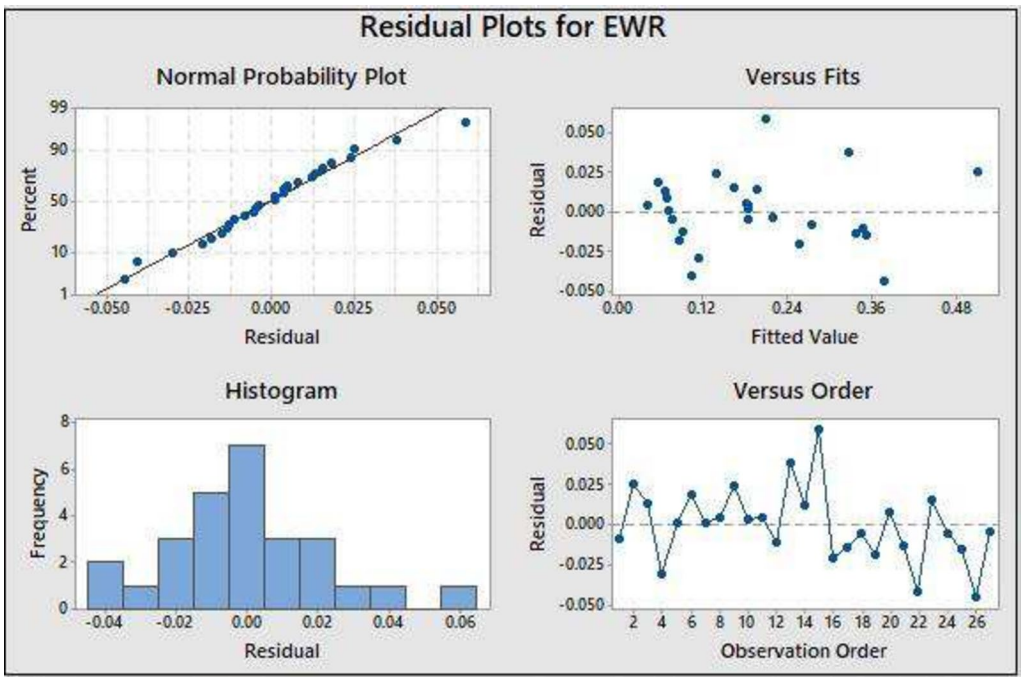

Fig. 5. Residual plots of RSM model for EWR.

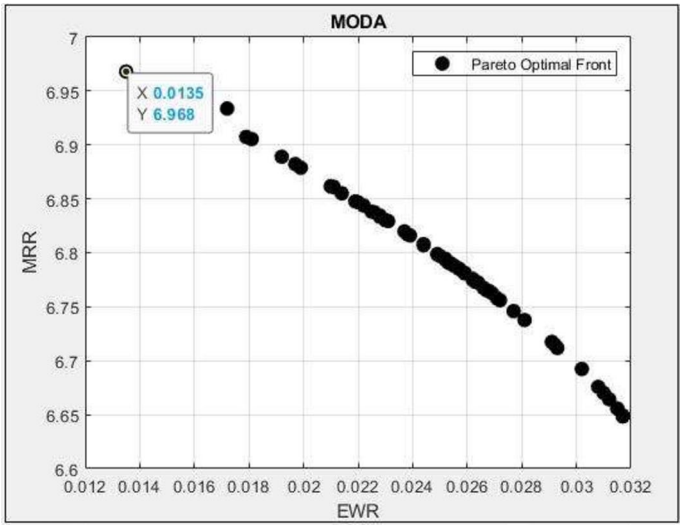

Fig. 6. Pareto-optimal frontier chart.

$$6 \leq \text{Ip} \leq 10$$
$$100 \leq \text{Ton} \leq 200$$
$$12 \leq \text{Tau} \leq 16$$
$$40 \leq \text{V} \leq 50$$

The concept of Pareto dominance was utilized to identify a set of non-dominated solutions[36,37]. Figure 6 illustrates the Pareto-optimal frontier, showing the distribution of points generated from the response optimization. Each point on the curve represents an optimal solution, giving users the flexibility to choose any point to conduct experiment. The non-dominated solutions (Pareto front) obtained from the MODA outputs were employed in the TOPSIS method to determine the most desirable solution. Equal importance was provided to each of the objectives. The best optimal solution was derived by executing the TOPSIS methodology. The point, highlighted on the Pareto frontiers (Fig. 6), signifies the optimal point selected through the TOPSIS technique. Table 7 presents the optimal values of the objective functions and their corresponding optimal decision parameters, obtained through the MODA-TOPSIS technique.

Experimental confirmation of optimal results

The confirmation test was performed with the MODA-TOPSIS predicted optimal input parameter setting (Table 7). The experimental results were compared with the predicted ones. The findings from the confirmation test and error percentages are shown in Table 8. It was observed that the error percentages for EWR and MRR stand at 6.67% and 2.54%, respectively. The validation of the MODA prediction model was confirmed by the lower error percentages observed between predicted and experimental values in EDM operation. Thus, it can be inferred that the experimental results validate the efficacy of the MODA-TOPSIS technique, making it a reliable method for predicting optimal process parameters in EDM operation.

Ip (Amps)	Ton (μs)	Tau	SV	MRR (mm^3/min)	EWR (mm^3/min)
6	200	12.00	41.60	6.9680	0.0135

Table 7. Results of optimization for MODA-TOPSIS techniques.

Responses	Predicted	Experimental	Error (%)
EWR (mm^3/min)	0.0135	0.0144	6.67
MRR (mm^3/min)	6.968	7.145	2.54

Table 8. Experimental validation of results.

Conclusion

This study focused on the Electrical Discharge Machining (EDM) processes applied to Monel K-500 Alloy by exploring various process parameters. An experimental design matrix was prepared using a Box-Behnken-based experimental array. The experiments were conducted according to this design matrix, and the resulting experimental dataset was used to develop a second-order polynomial regression model. The adequacy of this model was confirmed through ANOVA analysis, which demonstrated its high accuracy in predicting both the Material Removal Rate (MRR) and the Electrode Wear Rate (EWR).

The optimization of EDM process parameters was performed using the Multi-Objective Dragonfly Algorithm (MODA), relying on fitness functions to identify optimal solutions. In this optimization problem, EWR and MRR were treated as objective functions, with EWR being minimized and MRR maximized. Non-dominated optimal solutions for EWR and MRR were obtained, ranging from (0.0135, 6.968) to (0.0317, 6.649) along the Pareto optimal frontier. The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) was employed to select the most desirable optimal solution, assigning equal importance to each objective function. The optimal results for EWR and MRR in the EDM operation were found to be 0.0135 mm^3/min and 6.968 mm^3/min, respectively. These results were achieved with the following specific process parameters: pulse current (Ip) of 6 amps, pulse-on time (Ton) of 200 µs, duty cycle (Tau) of 12, and servo voltage (SV) of 41.6 V. The outcomes obtained through the MODA-TOPSIS method were validated with a confirmation test, demonstrating satisfactory performance.

Selecting these optimal process parameters can enhance the quality and efficiency of EDM processes involving Monel K-500. Although this study did not investigate the microstructure and surface integrity during EDM, this presents a promising area for future research.

Data availability

The data presented in this study are available in the article.

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Competing interests
The authors declare no competing interests.

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